

HSIE Dashboard Requirements and Design – User Guide

Hume Rural Health Alliance (HRHA)

Health Technology Services – eHealth

OFFICIAL



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Approvals

This section provides sign-off that all parties are aware of the information contained in this document.

*** All parties to complete this section*

Position	Name	Signature / Date (or email approval date)
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HUME CIO		
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This section specifies Agency contacts and details.

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Related Documents

Document
HL7 Unified Specification
DHHS HUME HSIE Dashboard User Guide

Definitions and Acronyms

ADT	Admission, Discharge and Transfer. It stores valuable patient information like the medical record numbers, ages, names, etc.
Agency	The Hume Rural Health Alliance
AIE	Agency Integration Engine (not to be confused with HRAIE). AIE references another integration engine downstream of the HRAIE.
ASCII	American Standard Code for Information Interchange. This is a character encoding standard for electronic communication.
DH	Department of Health
DR	Disaster Recovery environment
HTS	Health Technology Services
EMR	Electronic Medical Record
HIPS	Healthcare Information Provider Service. This is a middleware that sends data to my health records.
HL7	Health Level 7 Message Standard
HRAIE	Hume Region Alliance Agency Integration Engine. This is the one that Hume purchased.
HRHA	Hume Rural Health Alliance
HSIE	Health System Integration Engine. It standardises data flow across disparate clinical data systems. This is representative of a fleet.
iIE	i.PM Integration Engine
iPM	Patient management/administration system.
NON-PROD	Non-Production environment
PCMS	Patient and Client Management System used interchangeably with PAS – Patient Administration System
PROD	Production environment.
UR	Unique Record number.

RHEMS	Realtime Health Emergency Management System. This system has been replaced by Syndromic Surveillance.
SDLC	Systems Development Life Cycle
SIT	Systems Integration Testing
Vendor	Third party vendor such as Phillips, Dorevitch.

Intended Audience

The intended audience for this document includes:

- Office of Health Technology Service
- Integration Architecture and Development staff
- Key Health stakeholders

Introduction

Note: Throughout this document:

- The RHAPSODY application being monitored will be referred to as the “target” or “target system”. The box being used for RHAPSODY is the “target box”. There may be multiple target boxes.
- The web-based server application which does the actual monitoring of the target box is the “monitor” or “monitor server”. The box being used for this monitor is the “monitor box”.
- The user viewing pages presented by the monitor server on a workstation is the “client”. At any point in time there may be more than one client (i.e. the monitoring may be carried out at more than one location).

The HSIE Dashboard is web-based application which provides a status check of a RHAPSODY Integration Engine. To provide independent monitoring, it is implemented on a separate box to that used for the integration engine (i.e. if the target box goes down the monitoring remains active).

HUME Dashboard Architecture Illustration

The following illustrations depict the current state of the HUME interfaces and will be used as a basis for the HUME Dashboard. There are 3 sections to the interfaces.

- Input
- Processing (Not Required)
- Output

All these sections will be required to be viewable from the Dashboard. Messaging should be obtainable or viewable from the Dashboard.

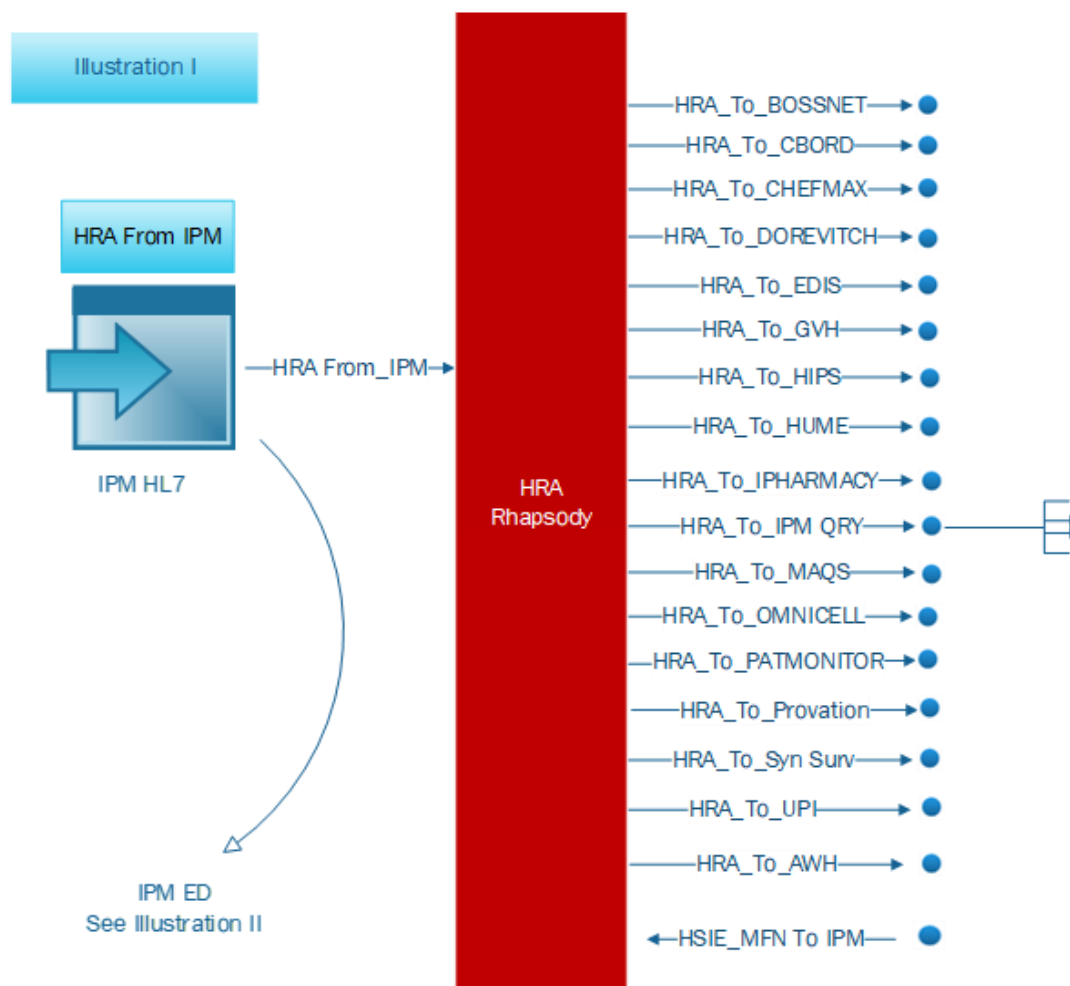


Illustration II

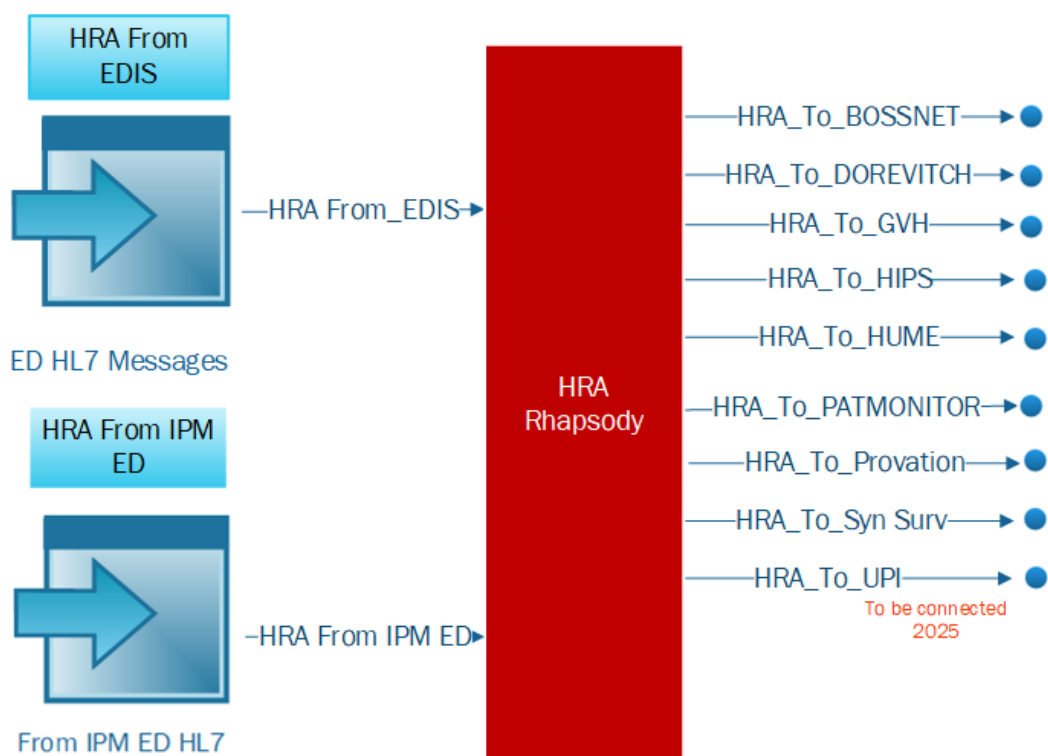


Illustration III

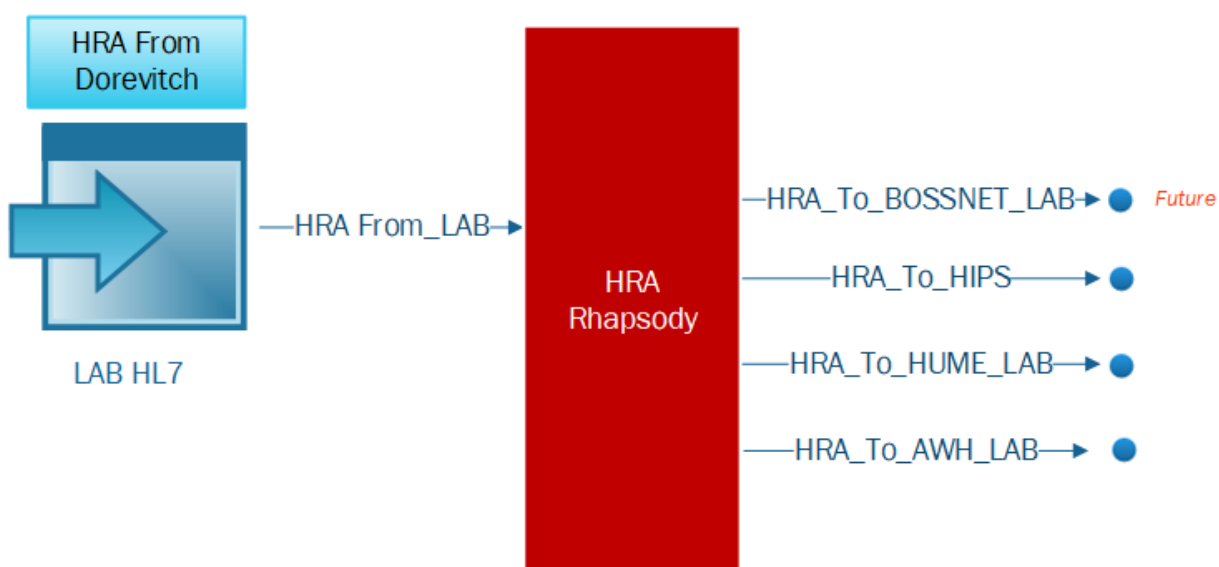
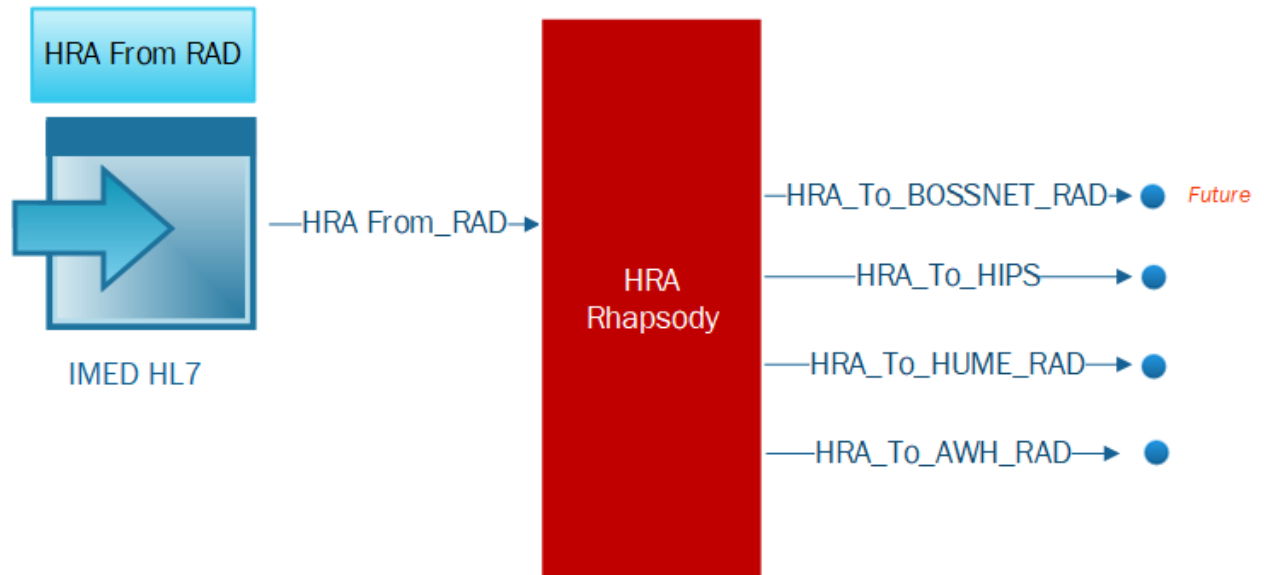


Illustration IIII

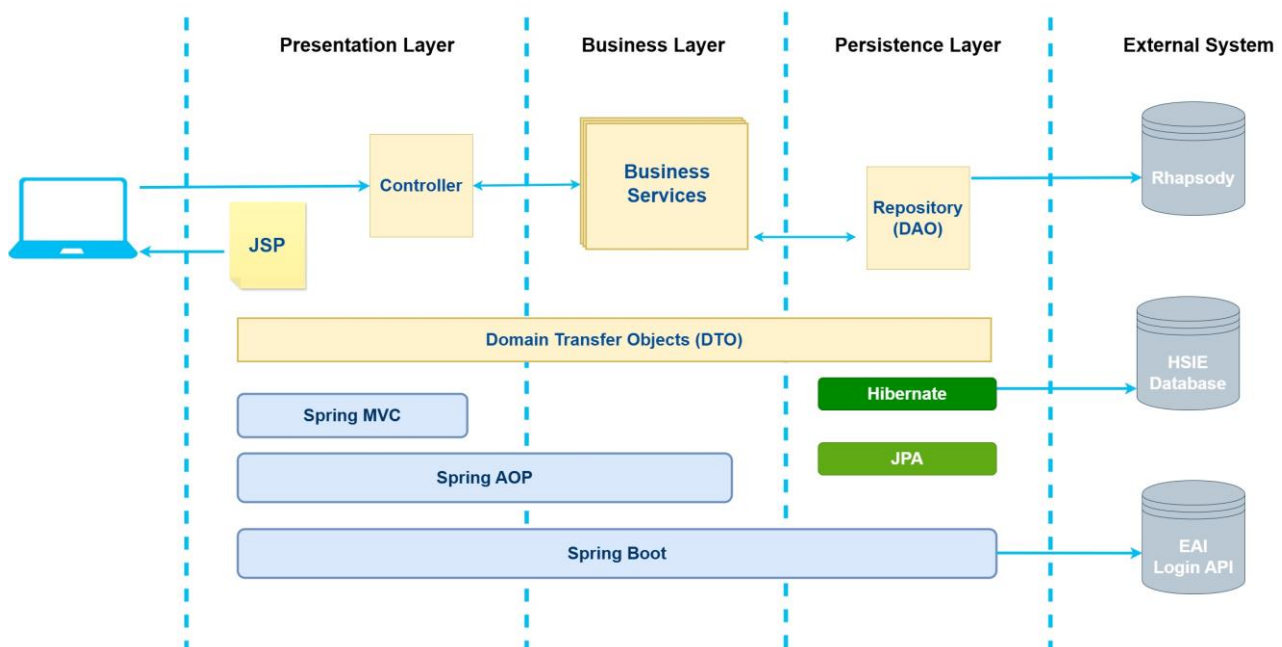


Architecture Components Diagram

The Spring Framework is designed with a layered architecture, meaning it consists of multiple interconnected modules. At its foundation lies the core container, which serves as the backbone for the entire framework.

Each module in the Spring framework is built on top of this core and provides specific functionalities that make enterprise application development easier and more efficient. These modules handle everything from dependency injection and web applications to security, data access, and seamless integration, allowing developers to choose and use only what they need for their projects.

Figure 1: The Spring Framework architecture



Packages and Directory Structure

The root package name starts 'au.org.htsv-dashbaord' and can be extended based on the module.

For example, in the case of controller classes, they are defined under 'au.org.htsv.dashboard.controller' package.

The project is organised into the following directories:

- src/ - root directory contains all of development codes
- src/main/ - The deployable source code of the project
- src/main/java – Java source code will be compiled when deploy requires
- src/main/resources – Supporting resources
- src/main/resources/static/css – Cascade Style Sheets
- src/main/resources/static/img – Image Files
- src/main/resources/static/js – Javascript Files
- src/main/resources/static/vendor – Vendor Javascript Framework Files
- src/main/resources/application.properties – Spring + Web configuration information
- src/main/webapp/WEB-INF/jps – View template components like JSP reside
- pom.xml – Address all dependencies of libraries in the project

Below shows the basic directory structure, although, this may slightly change in the future if required:

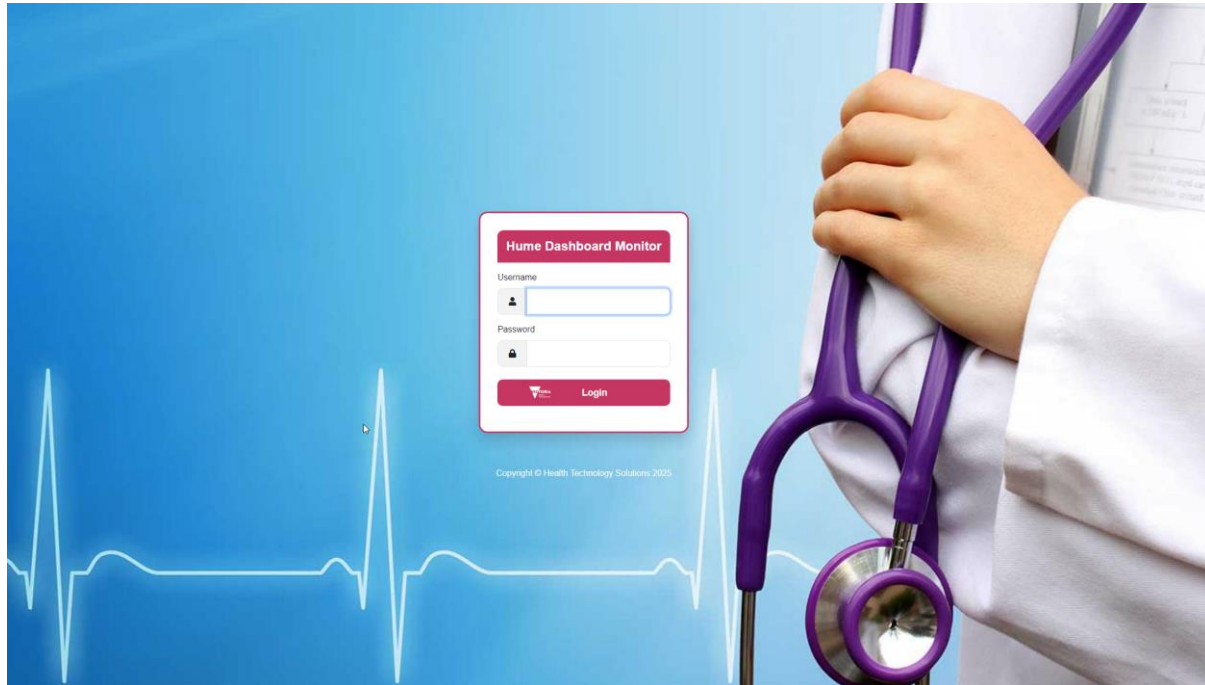
```

HumeDashboard [boot] [devtools] [zion master 111]
├── src/main/java
│   ├── au.org.hts.dashboard
│   ├── au.org.hts.dashboard.controller
│   ├── au.org.hts.dashboard.dao
│   ├── au.org.hts.dashboard.entity
│   ├── au.org.hts.dashboard.security
│   ├── au.org.hts.dashboard.service
│   └── au.org.hts.dashboard.util
├── src/main/resources
│   ├── static
│   │   ├── css
│   │   ├── img
│   │   ├── js
│   │   └── vendor
│   └── application.properties
├── JRE System Library [JavaSE-1.8]
├── Maven Dependencies
├── target/generated-sources/annotations
├── target/generated-test-sources/test-annotations
├── bin
├── lib
├── > src
├── target
│   ├── mvnw
│   ├── mvnw.cmd
│   └── pom.xml

```

Log In Screen

To Log In, users can use their existing HIPS Exception Report account credentials to log into the Dashboard portal.

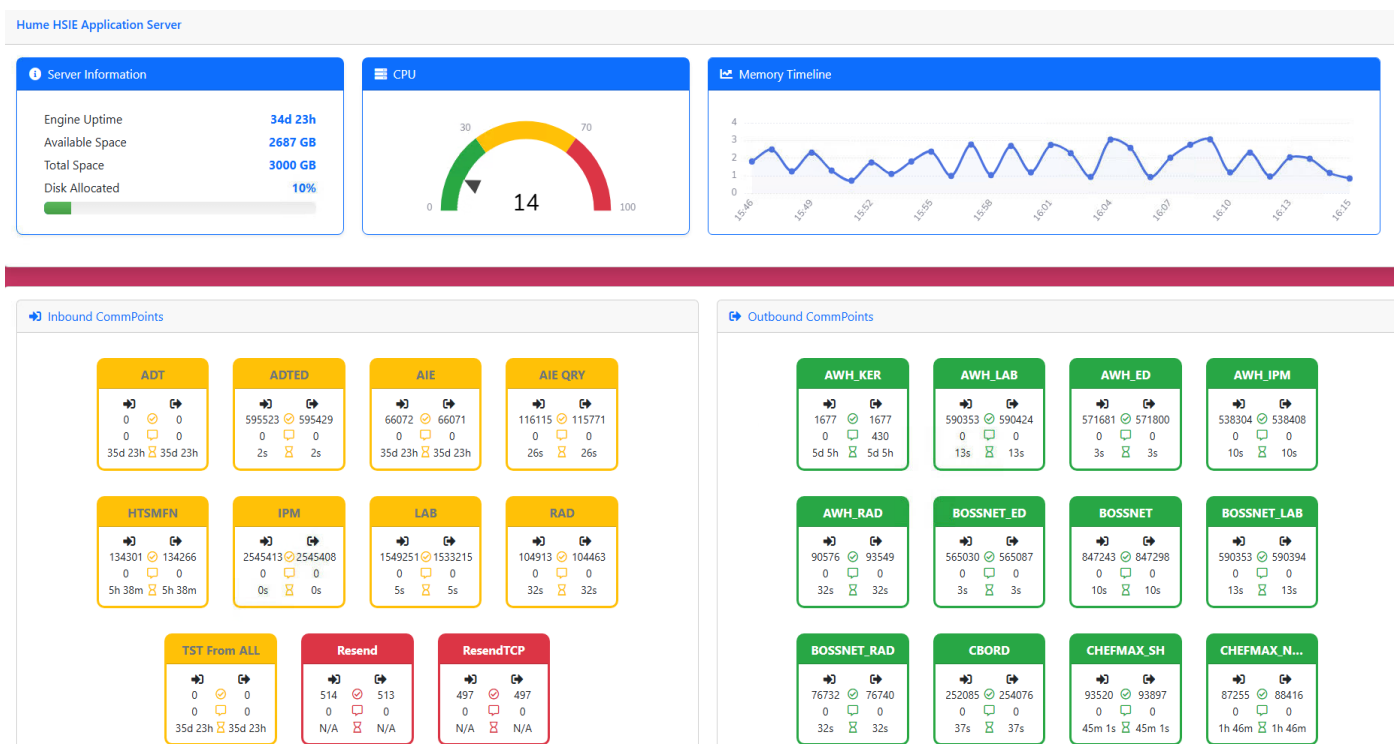


Dashboard User Interface (UI)

The dashboard landing page provides a visual representation of both Inbound and Outbound CommPoints. It features a dynamic layout that automatically adjusts to the user's screen resolution.

Inbound CommPoints represent interfaces that receive messages from external systems, while Outbound CommPoints represent interfaces that send messages to external systems. These are displayed on the left and right sides of the screen, respectively. Internal CommPoints points to Rhapsody Engine 6.

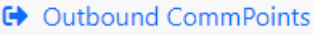
Figure 2. Shows an overview of the landing page of the dashboard.



Note: There are more Outbound CommPoints than Inbound CommPoints due to the Rhapsody connections. If a CommPoint is not visible on the screen, it is not up status.

Each interface group is visually represented, with the status of each interface indicated as outlined in **Table 1**. When hovering over a CommPoint box, users can view the full interface name and additional details about the messages being transmitted.

Table 1: Visual representations used within a CommPoint box and their meanings.

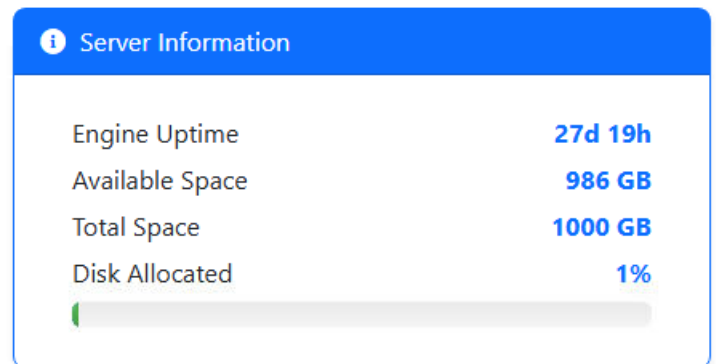
Icon Description	Visible Representation	Icon Definition and Meaning
Inbound CommPoints		The integration engine “Inbound CommPoints” interfaces receive messages from external systems.
Outbound CommPoints		The “Outbound CommPoints” interfaces send messages to external systems.
Green Comm Point Box		Indicates that CommPoint is functioning normally and is currently established more than one connection.
Orange Comm Point Box		Indicates that CommPoint is up but not currently connected. While the interfaces are running, they are awaiting a connection.
Red Comm Point Box		Indicates that CommPoint is down and may need to be re/started.
Incoming and Outcoming Messages		Indicates the directions showing incoming or outgoing.
Processed Messages		The numbers next to the icon represents the count of incoming or outgoing processed messages, respectively.
Current Message Count		The numbers next to the icon represents the number of queuing incoming or outgoing processed messages, respectively. Treated messages refer to those that have not been consumed and are currently queued.
Idle Time since Last Message		The numbers next to the icon indicates the idle time since the last consumed message, displayed in days, hours, minutes, and/or seconds.
Log Details		This is the third section of the UI, where users can click on an interface CommPoint box to view log file contents associated with the most recently selected interface.

Note: Hovering over titles, icons, or text will display additional details in a pop-up for more information.

Integration Engine Health Information

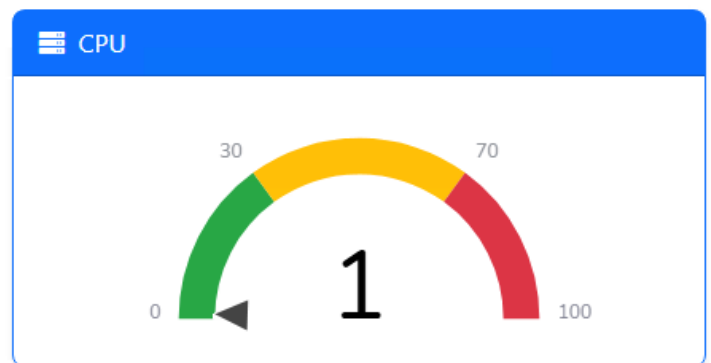
Server Information:

- **Engine uptime:** The total amount of time a Rhapsody engine has been continuously running without interruption.
- **Available Space:** This is the remaining unused storage currently available on that volume.
- **Total Space:** Refer to disk storage capacity on the server where the engine is running. This is the maximum disk capacity of the storage volume where the engine is installed or logging data.
- **Disk Allocated (Bar Graph):**
 - **Free Disk (White Bar Graph)** - Shows the available Free Disc space and percentage disc space in use on the target system.
 - **Percent Disc Usage (Green Bar Graph)** – Bar graph indicating usage of the available disc space (>80% usage is indicated in Red).



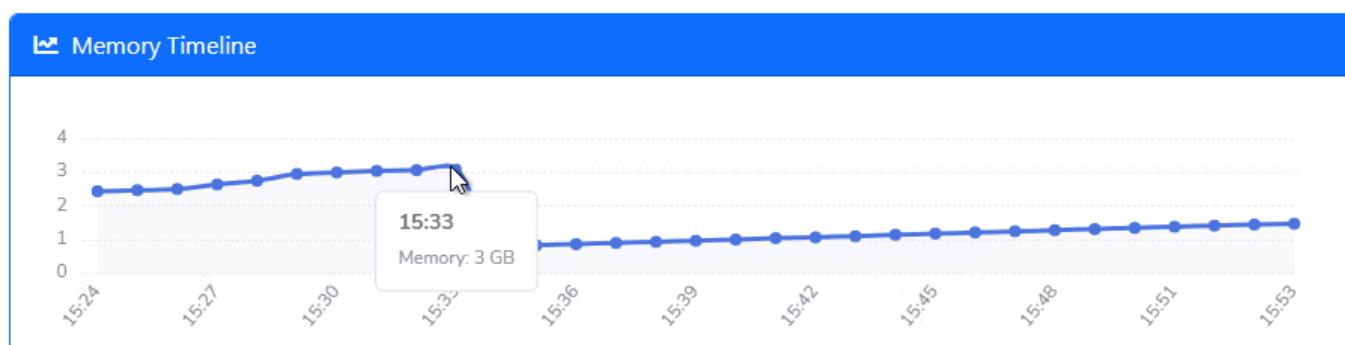
CPU (Central Processing Unit): Refers to how much processing power is currently being used. This uses a traffic light system.

- The **Green Zone** indicates use of **0 – 30 %** of the processing power. Nil action is needed.
- Sudden spikes into the **Yellow Zone (31 – 70% of the processing power)** may indicate:
 - A job is overloading the system
 - Delays in message processing or user commands
 - This zone may require attention.
- Where consistent high usage (e.g. **>70%** of the processing power - will be indicated by the **Red Zone**) this may also indicate:
 - Overloaded system
 - A process that's consuming too much CPU
 - This zone will require attention and potential action.



Memory Timeline graph shows the amount of memory (RAM) used over time on the integration engine. The graph's information is as follows:

- **X-axis (Horizontal):** Time (e.g. from 15:24 to 15:53), this is set as default for the last 30 minutes since the user logged in or when the user has refreshed the dashboard UI.
- **Y-axis (Vertical):** Memory usage in gigabytes (GB)
- **Blue line with dots:** Memory consumption at a specific time interval (i.e. the last 30 minutes since user logged in or when the user has refreshed the dashboard UI).
- **Tooltip (Hovered at time stamp of 15:33):** Shows that memory usage was 3 GB at that point in time.



Generally, there will be slight fluctuations in the Blue Line (Memory), this is normal. When the memory spikes as high as 4 – 7 GB, this will require attention.

Log Details

The CommPoint log is displayed at the bottom of the screen when a user clicks on a specific CommPoint box icon. There may be a short delay while the log is retrieved from the target system, after which the latest 10 log entries will appear. The scroll bar on the right allows users to navigate through the log area. Please note that the log display is a static snapshot and does not update in real time. To view the most recent data, click the "Refresh" button. If the session times out (after 15 minutes of inactivity), users will need to log back into the dashboard.

The screenshot displays a dashboard with a grid of CommPoint boxes at the top and a detailed log view below. The grid includes boxes for MANAD_BHS, MANAD_BH, MANAD_THS, MANAD_SH, MANAD_NHW, BOSSNET_ED (highlighted), BOSSNET, and HIPS. Each box shows a status icon, a value, and a refresh button. The log view, titled "Log Details for HRA To BOSSNET_ED_OI", shows a table of 10 log entries. A red arrow points from the BOSSNET_ED box to the log title. A scroll bar is visible on the right side of the log table.

DATE TIME	SENDING APP	RECEIVING APP	MSG TYPE	MSG EVENT	MSG CONT ID	PATIENT UR	GIVEN NAME	FAMILY NAME	GENDER	VISIT ID	HL7 MSG
14/05/2025 09:10:37	IPM	HSIE	ADT	A31	183846861				♂		☑
14/05/2025 08:29:31	IPM	HSIE	ADT	A04	183846860				♂		☑
14/05/2025 08:29:29	IPM	HSIE	ADT	A31	183846859				♂		☑
14/05/2025 08:22:46	IPM	HSIE	ADT	A31	183846858				♂		☑
13/05/2025 15:29:55	IPM	HSIE	ADT	A40	183846855				♂		☑
13/05/2025 15:28:56	IPM	HSIE	ADT	A40	183846855				♂		☑
13/05/2025 15:28:55	IPM	HSIE	ADT	A40	183846854				♂		☑
13/05/2025 15:27:55	IPM	HSIE	ADT	A40	183846854				♂		☑
13/05/2025 15:27:45	IPM	HSIE	ADT	A40	183846853				♂		☑
13/05/2025 15:26:46	IPM	HSIE	ADT	A40	183846853				♂		☑

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Table 2: Log Details box table titles and their meanings.

Icon Description	Icon Definition and Meaning
Date Time	The Date and Time the log file was sent.
Sending Application	The application sending the outgoing message. (i.e. Field MSH-3)
Receiving Application	The application receiving the incoming message. (i.e. Field MSH-5)
MSG Type	The type of HL7 message being sent. (i.e. Field MSH-9-1)
MSG Event	The type of trigger event of HL7 message being sent. (i.e. Field MSH-9-2)
MSG CONT ID	The unique ID number of the HL7 message. (i.e. Field MSH-10, also known as the Message Control ID)
Patient UR	The unique patient number within the HL7 message. (Can be found in field PID-3).
Given Name	The registered first name of the patient identified in the HL7 message. (Can be found in field PID-3).
Family Name	The registered surname of the patient identified in the HL7 message. (Can be found in field PID-3).
Gender	 = Male  = Female
Visit ID	The unique visit number of the HL7 message. (Can be found in field PV1-19).
HL7 MSG	 Users to click onto this icon, and the original HL7 message sent will pop up and display in a separate window. See figure below.



The screenshot shows a web application interface with a modal window titled "Original HL7 Message". The modal displays the following HL7 message text:

```
MSH|^~\&|HRA_EDIS|1650|HRA_BOSSNET|1650|20250512130842||ADT^A03|1000951|D|2.4
EVN|A03|20250512130842
PID|1||00000^HRA^1650||^HR^L||1987||M||9^PATIENT REFUSED TO
ANSWER|^VIC^3690^AUSTRALIA^4||^PRN^CP||1201|5|NSP|N
PV1|1|E|CORR^1650^CORRIDOR|1|2025000012||DOCT^DOCTOR^TEST|GP0001194171^
*RETURNING TO USUAL RE|||20250512125400|20250512130800
PV2||A02^ABNORMAL WEIGHT GAIN|07^PATIENT PREFERENCE||1|||ABNORMAL WEIGHT GAIN ALONG WITH RASHES ON THE
NECK|24||||N|||2

===== Ack =====

MSH|^~\&|HRA_EDIS|1650|HRA_BOSSNET|1650|20250512130842||ACK^A03|1000951|D|2.4
MSA|AA|1000951
```

Below the message text, there is an "Ack" section with the same MSH and MSA fields. The modal has a "Close" button at the bottom right. In the background, a table titled "Log Details for HRA To BOSSNET_ED_OI" is visible, showing columns for DATE TIME, SENDING APP, RECEIVING APP, MSG TYPE, MSG EVENT, MSG CONT ID, PATIENT UR, GIVEN NAME, FAMILY NAME, GENDER, VISIT ID, and HL7 MSG. The last row in the table has a red box around the "HL7 MSG" column header, which contains an envelope icon.